MATHIA

# PRIJEMNI

## Formule — Sve formule za prijemni iz matematike

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## I DEO Algebra, stepeni, koreni, logaritmi

### Stepeni i koreni

$$a^m a^n = a^{m+n} \qquad \frac{a^m}{a^n} = a^{m-n} \qquad (a^m)^n = a^{mn} \qquad a^{m/n} = \sqrt[n]{a^m} \qquad a^{-n} = \frac{1}{a^n}$$

### Logaritmi

$$\log_a(xy) = \log_a x + \log_a y \qquad \log_a \frac{x}{y} = \log_a x - \log_a y \qquad \log_a x^k = k \log_a x$$

$$\log_a x = \frac{\log_b x}{\log_b a} \qquad a^{\log_a x} = x \qquad \log_a a = 1, \quad \log_a 1 = 0$$

### Skraćeno množenje i apsolutna vrednost

$$(a \pm b)^2 = a^2 \pm 2ab + b^2 \qquad (a \pm b)^3 = a^3 \pm 3a^2b + 3ab^2 \pm b^3$$

$$a^2 - b^2 = (a-b)(a+b) \qquad a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$$

$$|x| < c \iff -c < x < c \qquad |x| > c \iff x < -c \lor x > c$$

## II DEO Jednačine i nejednačine

### Kvadratna jednačina

$$ax^2 + bx + c = 0 \Rightarrow x_{1,2} = \frac{-b \pm \sqrt{D}}{2a}, \quad D = b^2 - 4ac$$

$$x_1 + x_2 = -\frac{b}{a} \qquad x_1 x_2 = \frac{c}{a} \qquad ax^2 + bx + c = a(x - x_1)(x - x_2)$$

### Eksponencijalne i logaritamske

$$a^{f(x)} = a^{g(x)} \iff f(x) = g(x) \quad (a > 0,\ a \neq 1) \qquad \log_a f = \log_a g \iff f = g > 0$$

## III DEO Funkcije i nizovi

### Linearna i kvadratna funkcija

$$y = kx + n, \quad k = \tan\alpha = \frac{y_2 - y_1}{x_2 - x_1} \qquad y = ax^2 + bx + c: \quad x_T = -\frac{b}{2a}, \quad y_T = -\frac{D}{4a}$$

### Aritmetički niz

$$a_n = a_1 + (n-1)d \qquad S_n = \frac{n}{2}(a_1 + a_n) = \frac{n}{2}\left[2a_1 + (n-1)d\right]$$

## Geometrijski niz

$$a_n = a_1 q^{n-1} \qquad S_n = a_1\frac{q^n-1}{q-1}\ (q\neq 1) \qquad |q|<1:\ S=\frac{a_1}{1-q}$$

# IV DEO Trigonometrija

## Osnovni identiteti

$$\sin^2\alpha + \cos^2\alpha = 1 \qquad \tan\alpha = \frac{\sin\alpha}{\cos\alpha} \qquad 1+\tan^2\alpha = \frac{1}{\cos^2\alpha}$$

## Adicione formule i dvostruki ugao

$$\sin(\alpha\pm\beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta \qquad \cos(\alpha\pm\beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$$

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha = 1 - 2\sin^2\alpha$$

## Sinusna i kosinusna teorema

$$\frac{a}{\sin\alpha} = \frac{b}{\sin\beta} = \frac{c}{\sin\gamma} = 2R \qquad a^2 = b^2 + c^2 - 2bc\cos\alpha$$

# V DEO Geometrija i analitička geometrija

## Površine ravnih figura

$$P_{\triangle} = \frac{a\,h_a}{2} = \frac{1}{2}ab\sin\gamma \qquad P_{\square} = ab \qquad P_{krug} = r^2\pi,\ \ O = 2r\pi$$

$$\text{Heron:}\ \ P = \sqrt{s(s-a)(s-b)(s-c)}, \qquad s = \frac{a+b+c}{2}$$

## Analitička geometrija u ravni

$$d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2} \qquad k = \frac{y_2-y_1}{x_2-x_1} \qquad y - y_1 = k(x-x_1)$$

$$d(M,p) = \frac{|Ax_0+By_0+C|}{\sqrt{A^2+B^2}} \qquad (x-p)^2+(y-q)^2 = r^2$$

# VI DEO Kombinatorika i binomni obrazac

## Prebrojavanje

$$P_n = n! \qquad V_n^k = \frac{n!}{(n-k)!} \qquad C_n^k = \binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

## Binomni obrazac

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k \qquad \binom{n}{k} = \binom{n}{n-k}$$